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Solution of the congruence $x^m - 1 \equiv 0$
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Analysis of residues. Chapter six. First part.

237.

In Chapter III we showed that the congruence $x^n \equiv 1$, when the modulus is a prime number p , has μ roots when μ is the greatest common divisor of n and $p - 1$, and that these roots coincide exactly with the roots of the congruence $x^\mu \equiv 1$. It is therefore enough to consider the case in which n is a divisor of $p - 1$. Moreover, that the solution not only of the congruence $x^n - 1 \equiv 0$ but of any other congruence, for arbitrary moduli, can be derived from the solution for prime moduli has already been shown in various places and will be taught more fully below, in Chapter VIII.

238.

Nor is it necessary to stop here. In the same Chapter III we showed that the solution of $x^n \equiv 1$ depends on the resolution of similar congruences $x^a \equiv 1$, $x^b \equiv 1$, etc., where a , b , etc. are primes or powers of primes and n is their product. Namely, if A , B , etc. are arbitrary respective roots of $x^a \equiv 1$, $x^b \equiv 1$, etc., then the product $AB \dots$ will be one of the roots of $x^n \equiv 1$. Our investigations will therefore be restricted to the solution of $x^n \equiv 1 \pmod{p}$, where p is prime, n is a prime or a power of a prime, and at the same time n is a divisor of $p - 1$.

239.

It follows further from Chapter III that among the roots of $x^n \equiv 1$ there are always some through whose powers all the others can be exhibited. Thus, if r denotes such a root (we called it *primitive* above when $n = p - 1$, and we shall retain this expression here, although in a broader sense), all roots of the proposed congruence will be

$$1, \quad r, \quad rr, \quad r^3, \quad \dots, \quad r^{n-1}.$$

Such roots must therefore be sought with particular care, since once they have been found the remaining roots become evident of themselves. For brevity we denote any power of r by enclosing its exponent in parentheses, so that (0) denotes unity, (1) denotes any primitive root of $x^n \equiv 1$, (2) the square of (1), and so on. Thus the series (0), (1), (2), (3), $\dots$, $(n - 1)$ contains

all the roots. It is known, moreover, that (k) will always be such a primitive root whenever k is prime to n ; that is, in our case, where n is a power t^ν of a prime t , whenever t does not divide k . The signs (1) , (2) , etc. are plainly indeterminate in themselves; but once a determinate value has been assigned to (1) , all the others become determined.

240.

Since the aim is to investigate primitive roots before the others, they must first be separated from the rest. This is done by removing from the series $(0), (1), (2) \dots (n-1)$ all terms (k) for which k is divisible by t ; but if n is prime, or $\nu = 1$, only (0) is removed. Before proceeding to the investigation of the remaining roots, we urge the reader to make several examples, so that everything which without examples might perhaps appear too general may be seen concretely. We add one example, but it will not on that account be superfluous to work out others independently.

Let $p = 29$ and $n = 7$. The seven roots of $x^7 \equiv 1 \pmod{29}$ are 1, 7, 16, 20, 23, 24, 25. Since n is prime, all these roots except 1 are primitive. If therefore $7 = (1)$, the signs have the following meanings:

$$\begin{array}{cccccc} (0) & (1) & (2) & (3) & (4) & (5) & (6) \\ 1 & 7 & 20 & 24 & 23 & 16 & 25. \end{array}$$

The reader will remember that the signs (n) and (0) , $(n+1)$ and (1) , and in general (a) and (b) , are equivalent whenever $a \equiv b \pmod{n}$.

241.

For our purpose, however, we must proceed in another way as well. We retain only those terms (k) for which k is not divisible by t ; their number is $\frac{t-1}{t} n = \lambda$. All these numbers, or numbers congruent to them modulo n , can be exhibited by the successive powers of some number. Let this number be ρ . Then all the primitive roots of $x^n \equiv 1$ are denoted by

$$(1), \quad (\rho), \quad (\rho^2), \quad (\rho^3), \quad \dots, \quad (\rho^{\lambda-1}).$$

By this device we obtain the complete exclusion of the non-primitive roots; the reasons and advantages of this will become clearer below. In our example we may put $\rho = 3$, and the primitive roots of $x^7 \equiv 1$ are arranged as

$$\begin{array}{cccccc} (1) & (3) & (3^2) & (3^3) & (3^4) & (3^5) \\ \text{that is } (1) & (3) & (2) & (6) & (4) & (5) \\ 7 & 24 & 20 & 25 & 23 & 16. \end{array}$$

242.

So that the reader may know where the following investigations are headed, it will be useful to state the theorem which we propose to prove and explain.

If the number λ , which is $t^{\nu-1}(t-1)$, has the distinct prime factors a, b, c, d , etc., and if $\lambda = a^\alpha b^\beta c^\gamma \dots$, then the resolution of the congruence $x^n - 1 \equiv 0$ depends on the resolution of $\alpha + \beta + \dots$ lower congruences, of which α are of degree a , β of degree b , γ of degree c , etc.

Thus in our example the resolution of $x^7 \equiv 1$ depends on one congruence of the second degree and another of the third degree; and in general it is clear that the degree of these congruences never depends on the modulus p . In order to arrive at the proof of this theorem, we must first put forward some propositions concerning the connection between congruences and their roots, although these investigations properly belong to Chapter VIII, where they will be pursued further.

243.

THEOREM. Suppose that the congruence

$$x^m + Ax^{m-1} + Bx^{m-2} + \dots + N \equiv 0 \pmod{\text{prime}}$$

is so constituted that, when the product of the m factors $x - r, x - r', x - r'', x - r''' \dots$ is formed, and is $x^m + ax^{m-1} + bx^{m-2} + \dots + n$, one has $A \equiv a, B \equiv b, C \equiv c$, etc. modulo p . Then the quantities $r, r', r'' \dots$ are roots of the proposed congruence, and it has no others.

Proof. I. We always have

$$x^m + Ax^{m-1} + Bx^{m-2} + \dots \equiv x^m + ax^{m-1} + bx^{m-2} + \dots \pmod{p}.$$

The latter expression becomes 0 on putting $x = r, x = r', x = r''$, etc.; hence, for these values of x , the former expression becomes 0 (mod p). This proves the first assertion.

II. If some further value ϱ , congruent to none of r, r' , etc., satisfied the proposed congruence, then

$$\begin{aligned} 0 &\equiv \varrho^m + A\varrho^{m-1} + B\varrho^{m-2} + \dots \\ &\equiv \varrho^m + a\varrho^{m-1} + b\varrho^{m-2} + \dots \\ &\equiv (\varrho - r)(\varrho - r')(\varrho - r'')(\varrho - r''') \dots \end{aligned}$$

But none of the factors $\varrho - r, \varrho - r', \varrho - r''$, etc. is 0 modulo p ; hence, since p is prime, it is absurd that their product should be 0. Therefore there are no roots besides r, r' , etc. This proves the second assertion.

244.

PROBLEM. Let $r, r', r'' \dots$ be unknown quantities whose number is m , whose sum is α , whose sum of squares is β , whose sum of cubes is γ , and so on, and whose sum of m -th powers is μ . Suppose that these numbers themselves, again m in number, are not given, but rather other numbers α', β', γ' , etc. respectively congruent to them modulo a prime $p > m$. Find the congruence of the m -th degree whose roots are r, r', r'' , etc.

Solution. Regard r, r', r'' , etc. as the roots of an equation

$$x^m + Ax^{m-1} + Bx^{m-2} + Cx^{m-3} + \dots = 0$$

and determine its coefficients A, B, C , etc., using congruence in place of equality, by the known method, namely by setting

$$\begin{aligned} -A &\equiv \alpha', \\ -2B &\equiv \beta' + A\alpha', \\ -3C &\equiv \gamma' + A\beta' + B\alpha', \\ -4D &\equiv \delta' + A\gamma' + B\beta' + C\alpha', \\ &\text{etc.} \\ -mN &\equiv \mu' + A\lambda' + \text{etc.} \end{aligned}$$

These coefficients cannot be indeterminate, since all the numbers $1, 2, 3 \dots m$ are less than p . I say that the required congruence is

$$x^m + Ax^{m-1} + Bx^{m-2} + \dots + N \equiv 0.$$

Proof. Suppose that the equation whose roots are r, r', r'', r''' , etc. is

$$x^m + ax^{m-1} + bx^{m-2} + \dots = 0.$$

Then

$$\begin{aligned} -a &= \alpha, \\ -2b &= \beta + a\alpha, \\ -3c &= \gamma + a\beta + b\alpha, \\ -4d &= \delta + a\gamma + b\beta + c\alpha, \\ &\text{etc.} \end{aligned}$$

It is therefore evident that

$$a \equiv A, \quad b \equiv B, \quad c \equiv C \quad \text{etc.} \quad (\text{mod } p).$$

Thus, by the preceding section, the numbers $r, r', r'',$ etc., which are roots of

$$x^m + ax^{m-1} + bx^{m-2} + \dots = 0,$$

are also roots of the congruence

$$x^m + Ax^{m-1} + Bx^{m-2} + \dots \equiv 0. \quad \text{Q. E. D.}$$

We leave examples for the readers to construct.

245.

We return to our purpose. Keeping the notation of §§242 and the preceding sections, we set out to show that, if λ is the product of any factors efg , etc., the primitive roots of $x^n \equiv 1$, whose number is λ , can be divided into e classes, so that the sums of the roots put into the same class are given by a congruence of degree e ; and, once these are supposed known, any class can be subdivided into f orders, so that the sums of the terms in each order are given by a congruence of degree f . These orders can then again be subdivided, and so on, until one arrives at the individual roots.

246.

Definition. We call the aggregate of all terms contained in a form $(\rho^{ke+\alpha})$ (§241) a *complete period*, or simply a *period*. Here e denotes a divisor of λ , α any given number, and k all integers from 0 up to $\frac{\lambda}{e} - 1$. For brevity we denote such a period by $(e * \alpha)$. Thus in our example the terms

$$\begin{array}{ll} (1), (2), (4) & \text{constitute the periods } (2 * 0), \\ (3), (6), (5) & (2 * 1), \\ \text{while } (1), (6) & \text{constitute } (3 * 0), \\ (3), (4) & (3 * 1), \\ (2), (5) & (3 * 2). \end{array}$$

If all the terms are distributed in any way into periods, and each period is again distributed into smaller periods, and so on, we say that what was promised in the preceding section has been obtained.

Before entering upon this exposition, however, we shall show that although the formation of such a period depends in a certain way on two arbitrary quantities r and ρ , it nevertheless contains nothing vague: however these quantities are chosen, the same terms always come together into the same period, provided that the number of terms which the period is to contain has been prescribed.

The criterion for two terms A and B to be in the same period is that both are contained in such a form $(\rho^{ke+\alpha})$, that is,

$$A \equiv r^{\rho^{ke+\alpha}}, \quad B \equiv r^{\rho^{le+\alpha}} \pmod{p}.$$

Here r is a primitive root of $x^n \equiv 1 \pmod{p}$, while ρ is a primitive root of $x^\lambda \equiv 1 \pmod{n}$, as above.

We must prove that if, in place of r and ρ , other numbers are chosen, say s and σ , then A and B are contained in similar forms $s^{\sigma^{ke+\theta}}$ and $s^{\sigma^{le+\theta}}$.

Let $s^m \equiv r \pmod{p}$, $\sigma^\mu \equiv \rho \pmod{n}$, and $m \equiv \sigma^\zeta \pmod{n}$. This can be done because r and ρ are primitive roots. Moreover m is prime to n , and μ to λ (Chapter III). By the appropriate substitutions we obtain

$$A \equiv s^{\sigma^{ke+\mu+\zeta}}, \quad B \equiv s^{\sigma^{le+\mu+\zeta}}. \quad \text{Q. E. D.}$$

247.

THEOREM. *The product of two similar periods can be composed, independently of the number p , by the addition of similar periods and given numbers.*

We call periods similar when they contain the same number of terms, or equivalently when the number e is the same.

Example. Let $n = 7$. The product of the periods $(1) + (6)$ and $(2) + (5)$ is, since $(a) \times (b) = (a + b)$, equal to $(3) + (6) + (8) + (11)$, that is, it consists of the periods $(3) + (4)$ and $(1) + (6)$.

Proof. Let $\frac{\lambda}{e} = f$, and let the given periods $(e * \alpha)$ and $(e * \delta)$, or sums, be

$$\begin{aligned}(\rho^\alpha) + (\rho^{\alpha+e}) + (\rho^{\alpha+2e}) + \dots + (\rho^{\alpha+(f-1)e}) &= P, \\(\rho^\delta) + (\rho^{\delta+e}) + (\rho^{\delta+2e}) + \dots + (\rho^{\delta+(f-1)e}) &= Q.\end{aligned}$$

The product PQ consists of f^2 terms. These must be arranged as follows. Form f series, each consisting of f terms. The first contains the product of P by (ρ^δ) , the second the product $P(\rho^{\delta+e})$, and so on. In the first series the first place is occupied by the product arising from the part (ρ^α) , the second by the product arising from $(\rho^{\alpha+e})$, and so on. In the second series the first place is assigned to the product arising from $(\rho^{\alpha+e})$, the second to the product arising from $(\rho^{\alpha+2e})$, and so on, the last to that arising from (ρ^α) . The third begins with the product arising from $(\rho^{\alpha+2e})$, and so forth; after the product from the last part follow the products from the first and second parts, etc. Thus, if the successive parts of the period P are denoted by $1, 2, 3 \dots z$ and those of the period Q by I, II, III, $\dots Z$, the parts of PQ are

$$\begin{aligned}1 \cdot \text{I} + 2 \cdot \text{I} + 3 \cdot \text{I} + 4 \cdot \text{I} + \dots + z \cdot \text{I}, \\2 \cdot \text{II} + 3 \cdot \text{II} + 4 \cdot \text{II} + \dots + 1 \cdot \text{II}, \\3 \cdot \text{III} + 4 \cdot \text{III} + \dots + 1 \cdot \text{III} + 2 \cdot \text{III}, \\\text{etc.}\end{aligned}$$

Now collect into f orders all terms occupying the same place in the several series. I say, first, that if one term is $\equiv 1$, then all the others of the same order are also $\equiv 1$; and second, that every order in which no term is $\equiv 1$ forms a period. Once these have been proved, the proposition is plainly obtained.

The general form of such an order is

$$(\rho^{\alpha+ke} + \rho^\delta), \quad (\rho^{\alpha+(k+1)e} + \rho^{\delta+e}), \quad (\rho^{\alpha+(k+2)e} + \rho^{\delta+2e}), \dots, (\rho^{\alpha+(k+f-1)e} + \rho^{\delta+(f-1)e}).$$

For $\rho^{\alpha+(k-1)e}$ one may also write $\rho^{\alpha+(k+f-1)e}$, since $ef = \lambda$ and $\rho^\lambda \equiv 1 \pmod{n}$, and similarly for the preceding terms. Put $\rho^{\alpha+ke} + \rho^\delta \equiv \rho^\chi \pmod{n}$, which is permitted unless perhaps $\rho^{\alpha+ke} + \rho^\delta$ is divisible by n .¹ Then the order can be exhibited as

$$(\rho^\chi), \quad (\rho^{\chi+e}), \quad (\rho^{\chi+2e}), \dots, (\rho^{\chi+(f-1)e}),$$

which is plainly the period $(e * \chi)$. If, on the other hand, $\rho^{\alpha+ke} + \rho^\delta$ is divisible by n , all terms of the order are $\equiv (0)$, that is, $\equiv 1$. Q. E. D.

Remark. This proof also gives a very easy method for expanding the product. We shall give another below which does not have this advantage, but which is not to be despised on account of its simplicity.

248.

All the smaller periods which constitute a larger period are called a system of periods. Thus the periods

$$(ef * \alpha), \quad (ef * f + \alpha), \quad (ef * 2f + \alpha), \dots, (ef * (e-1)f + \alpha)$$

¹The proposition must be expressed somewhat differently if n denotes, in general, a power of a prime; when n is prime, nothing need be changed.

which compose the period $(f * \alpha)$ are designated by this name. A system is *properly ordered* if the numbers placed after the sign $*$, here $\alpha, f + \alpha, 2f + \alpha$, proceed in an arithmetic progression with difference f ; and systems are *similar* if both the smaller periods and the larger periods are similar.

THEOREM. *If the periods of two similar properly ordered systems are multiplied with one another, the first by the first, the second by the second, the third by the third, etc., the sum of all these products can be composed of periods similar to the larger period and of given numbers.*

Proof. Let the systems be

$$\begin{aligned} (ef * \alpha), & (ef * \alpha + f), (ef * \alpha + 2f), \dots, \\ (ef * \delta), & (ef * \delta + f), (ef * \delta + 2f), \dots \end{aligned}$$

The products of each period of the first system with the corresponding periods of the second consist, by the preceding section, of integers and similar periods. A little attention to the genesis of these periods shows that if

$$(ef * \alpha) \times (ef * \delta)$$

consists of an integer N and of the periods $(ef * A), (ef * B), (ef * C)$, etc., then the products

$(ef * \alpha + f) \times (ef * \delta + f)$ consist of N and of $(ef * A + f), (ef * B + f), (ef * C + f)$, etc.,
 $(ef * \alpha + 2f) \times (ef * \delta + 2f)$ consist of N and of $(ef * A + 2f), (ef * B + 2f), (ef * C + 2f)$, etc.,

and in general

$(ef * \alpha + \mu f) \times (ef * \delta + \mu f)$ consists of N and of $(ef * A + \mu f), (ef * B + \mu f), (ef * C + \mu f)$, etc.

It follows at once that the sum of all the products is

$$eN + (f * A) + (f * B) + (f * C) \text{ etc.} \quad \text{Q. E. D.}$$

This proof also supplies a method for finding that sum.

249.

It is easy to make this theorem still more general. If any number of similar properly ordered systems are given and one forms the products of all first periods, of all second periods, etc., then the sum of all these products consists of numbers and larger periods. If all these systems are taken equal, the sum of any powers whatever of all the periods will consist of numbers and periods similar to the larger one. It is now clear what this is aiming at. Let $\lambda = efgh\dots$, and let all the first roots be split into e periods A, A', A'' , etc.; let each of these again be split into f periods B, B', B'' , etc.; let each of these be split into g periods C, C', C'' , etc. The sum of all the periods is already given, namely it is $\equiv -1$. But, according to what we have just said, the following are also given:

$$\begin{aligned} A^2 + (A')^2 + (A'')^2 + (A''')^2 + \text{etc.}, \\ A^3 + (A')^3 + (A'')^3 + (A''')^3 + \text{etc.}, \\ \text{etc.} \end{aligned}$$

Hence, by §244, a congruence of degree e can be found whose roots are $A, A', A'',$ etc. Once these are supposed known, any period is divided into smaller ones,

$$\begin{array}{lcl} A & \text{into} & B, \quad B', B'', \dots \\ A' & \text{into} & B^{(n)}, \quad B^{(n+1)}, B^{(n+2)}, \dots \\ & & \text{etc.} \end{array}$$

Thus $B + B' + B'' + \dots \equiv A$ is known. But it is known that

$$\begin{array}{l} B^2 + (B')^2 + (B'')^2 + \dots, \\ B^3 + (B')^3 + (B'')^3 + \dots, \\ \text{etc.} \end{array}$$

can be expressed from units and the periods $A, A', A'',$ etc. Therefore $B, B', B'',$ etc. are given by a congruence of degree f , from which they can be found. In the same way the periods of which $A', A'',$ etc. consist can be determined. From this anyone will see that by a completely similar method any period can be subdivided into smaller ones until the roots themselves are reached.

250.

In applying these rules, however, a difficulty occurs which must be removed. Since every congruence has several roots, we must see which sign is to be assigned to each so that they can be properly distinguished from one another. Since the designation of periods depends on the numbers r and ρ , which may be chosen arbitrarily, something arbitrary must necessarily cling to the designation as well. The number ρ must of course be fixed at the beginning. The essential character of our method consists chiefly in deriving smaller periods from larger ones. But this cannot be done without the proper order of the periods, which we have achieved by the signs. We must therefore strive to distinguish all periods by their signs as soon as they have been found.

Let the period A be designated by $(e * \alpha)$ and split into f periods $B, C, D,$ etc., which must be designated. It is clear that each is contained in a form $(ef * ke + \alpha)$; but I say that for one of them, say B , the number k may be taken arbitrarily, and the placement of the others can then be derived.

Let R be some primitive root of $x^n \equiv 1$, and suppose B consists of the terms $R^\mu + R^\nu + \dots$. Let $\frac{1}{\mu} \rho^{ke+\alpha} \equiv \frac{1}{\nu} \pmod{n}$. Since the value of r is arbitrary, provided only that A obtains the sign $(e * \alpha)$, which it is clear can be done, put $r \equiv R^\nu \pmod{p}$. Then the first term of B will be $r^{\rho^{ke+\alpha}}$, and B may be designated by $(ef * ke + \alpha)$. If, instead of the term R^μ , we had considered the term R^ν , we would have obtained another value of r ; but it is easily seen that for any root ρ , the root r can have $\frac{\lambda}{ef}$ different values.

251.

We now consider how the remaining periods are distinguished by their signs from the designation of one period. For this purpose another method must be sought for finding the remaining

periods, since, insofar as the remaining periods are, like A itself, roots of some congruence, no order is visible among them. Suppose that A is designated by $(ef * 0)$. From the preceding results it follows that

$$\begin{aligned} A^2 & \text{ has the form } M + N(ef * 0) + O(ef * 1) + P(ef * 2) + \cdots, \\ A^3 & \text{ has the form } M' + N'(ef * 0) + O'(ef * 1) + \cdots, \\ & \text{etc.} \\ A^{ef-1} & \text{ has the form } M^* + N^*(ef * 0) + O^*(ef * 1) + \cdots. \end{aligned}$$

To these is added the congruence

$$(ef * 0) + (ef * 1) + \cdots + (ef * ef - 1) \equiv -1.$$

We therefore have $ef - 1$ linear congruences and just as many unknown quantities, which can consequently be determined by elimination.

Remark. It may happen that the unknown quantities are given by such expressions as $\frac{V}{Wp}$. How this difficulty may be remedied will be taught below. Here, because this case can occur only very rarely, we shall not dwell on it.

252.

This is enough in general concerning the solution of pure congruences. Many further points about them will be said below in passing; in particular, much can be transferred here from the solution of pure equations, and we shall not neglect to note this in its proper place. We add one more example, which, when compared with the precepts, will make everything clearer to less experienced readers.

Let $n = 31$ and $p = 311$; that is, we must investigate the roots of

$$x^{31} - 1 \equiv 0 \pmod{311}.$$

First one must seek a primitive root of $y^{30} - 1 \equiv 0 \pmod{31}$; such a root is $y \equiv 3$. Put therefore $\rho \equiv 3$ and first divide all primitive roots of the proposed congruence into five periods, namely

$$\begin{aligned} (5 * 0) &= (1) + (26) + (25) + (30) + (5) + (6), \\ (5 * 1) &= (3) + (16) + (13) + (28) + (15) + (18), \\ (5 * 2) &= (9) + (17) + (8) + (22) + (14) + (23), \\ (5 * 3) &= (27) + (20) + (24) + (4) + (11) + (7), \\ (5 * 4) &= (19) + (29) + (10) + (12) + (2) + (21). \end{aligned}$$

By the required calculations one finds the sum of the periods to be $\equiv -1$, the sum of their squares $\equiv 25$, of their cubes $\equiv 26$, of their fourth powers $\equiv 249$, and of their fifth powers $\equiv 564$. Hence the periods are roots of the congruence

$$x^5 + x^4 - 12x^3 - 21x^2 + x + 5 \equiv 0.$$

Furthermore one finds

$$\begin{aligned} (5 * 0)^2 &\equiv 6 + 2(5 * 0) + 2(5 * 3) + (5 * 4), \\ (5 * 0)^3 &\equiv 12 + 15(5 * 0) + 4(5 * 1) + 3(5 * 2) + 6(5 * 3) + 6(5 * 4), \\ (5 * 0)^4 &\equiv 90 + 60(5 * 0) + 28(5 * 1) + 26(5 * 2) + 49(5 * 3) + 38(5 * 4), \end{aligned}$$

and hence, by elimination,

$$\begin{aligned}
5(5 * 1) &\equiv 3(5 * 0)^4 - (5 * 0)^3 - 33(5 * 0)^2 - 24(5 * 0) + 15, \\
5(5 * 2) &\equiv -2(5 * 0)^4 - (5 * 0)^3 + 22(5 * 0)^2 + 31(5 * 0), \\
5(5 * 3) &\equiv (5 * 0)^4 - 2(5 * 0)^3, \\
5(5 * 4) &\equiv -2(5 * 0)^4 + 4(5 * 0)^3.
\end{aligned}$$

One root of the congruence just found is 17; therefore, if $(5 * 0) \equiv 17$, then $(5 * 1) \equiv 183$, $(5 * 2) \equiv 263$, $(5 * 3) \equiv 91$, $(5 * 4) \equiv 67$.

Now each of the periods found is again split into triples, namely

$$\begin{aligned}
(5 * 0) &\text{ into } (15 * 0), (15 * 5), (15 * 10), \text{ or into } (1) + (30), (26) + (5), (25) + (6), \\
(5 * 1) &\text{ into } (15 * 1), (15 * 6), (15 * 11), \text{ or into } (3) + (28), (16) + (15), (13) + (18), \\
&\text{etc. etc.}
\end{aligned}$$

Suppose that the periods into which the following are divided are roots of the respective congruences

$$\begin{aligned}
(5 * 0) &\quad x^3 + Ax^2 + Bx + C \equiv 0, \\
(5 * 1) &\quad x^3 + A'x^2 + B'x + C' \equiv 0, \\
(5 * 2) &\quad x^3 + A''x^2 + B''x + C'' \equiv 0, \\
&\text{etc.}
\end{aligned}$$

Then

$$\begin{aligned}
A &\equiv -(5 * 0), & B &\equiv (5 * 0) + (5 * 3), & C &\equiv -2 - (5 * 4), \\
A' &\equiv -(5 * 1), & B' &\equiv (5 * 1) + (5 * 4), & C' &\equiv -2 - (5 * 0), \\
&\text{etc.} & &\text{etc.} & &\text{etc.}
\end{aligned}$$

Therefore

$$\begin{aligned}
(15 * 0), (15 * 5), (15 * 10) &\text{ are roots of } x^3 - 17x^2 + 108x - 60 \equiv 0, \\
(15 * 1), (15 * 6), (15 * 11), \\
(15 * 2), (15 * 7), (15 * 12), \\
(15 * 3), (15 * 8), (15 * 13), \\
(15 * 4), (15 * 9), (15 * 14).
\end{aligned}$$

Here one has

$$\begin{aligned}
(15 * 0)^3 - 3(15 * 0) &\equiv (15 * 1), \\
(15 * 1)^3 - 3(15 * 1) &\equiv (15 * 2), \\
&\text{etc.}
\end{aligned}$$

Hence, if one root of the first congruence, namely 10, is set equal to $(15 * 0)$, one obtains

$$\begin{aligned}
(15 * 0) &\equiv 10, & (15 * 5) &\equiv , & (15 * 10) &\equiv , \\
(15 * 1) &\equiv 37, & (15 * 6) &\equiv , & (15 * 11) &\equiv , \\
(15 * 2) &\equiv -151, & (15 * 7) &\equiv , & (15 * 12) &\equiv , \\
(15 * 3) &\equiv -39, & (15 * 8) &\equiv , & (15 * 13) &\equiv , \\
(15 * 4) &\equiv -112, & (15 * 9) &\equiv , & (15 * 14) &\equiv .
\end{aligned}$$

Finally, when the constituent terms of these individual periods are taken, we have

$$\begin{array}{ll} (1), (30) & \text{roots of } x^2 - (15 * 0)x + 1 \equiv 0, \\ (3), (28) & x^2 - (15 * 1)x + 1 \equiv 0, \\ & \text{etc.} \end{array}$$

The roots of the first congruence are 126 and 195, and these are therefore primitive roots of $x^{31} \equiv 1$; from them the remaining roots can be derived without difficulty.